# Solutions and separating

Locate and open the learning object, *Solutions explorer.* Select **Start** to display the screen titled 'What are the molecules doing in solution?'

1. Watch what happens when a sugar cube is dropped into the beaker of water. Describe what happens to the particles of sugar as it dissolves.

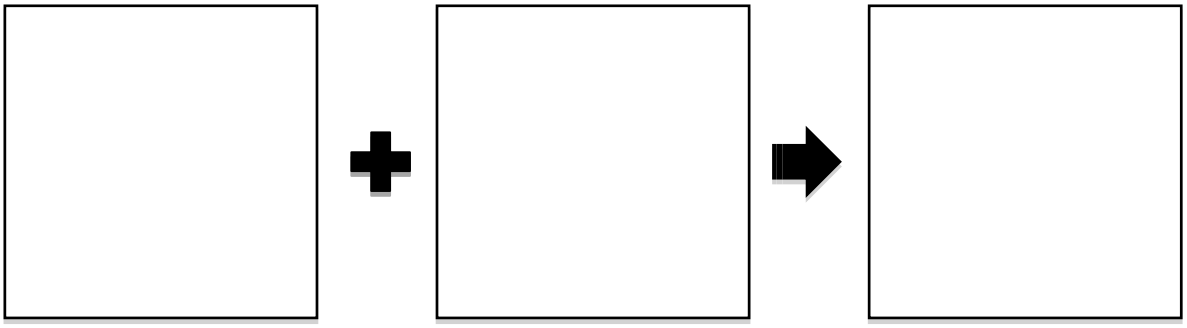
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2. Draw a particle model of the solid, liquid and solution.



solid liquid solution

3. Watch molecules of water bump into sugar. Why do you think only some collisions cause sugar molecules to break off the cube?

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4. In the animation, which substance is:

   the solute

   .....

   the solvent

   .....

   the solution

   .....

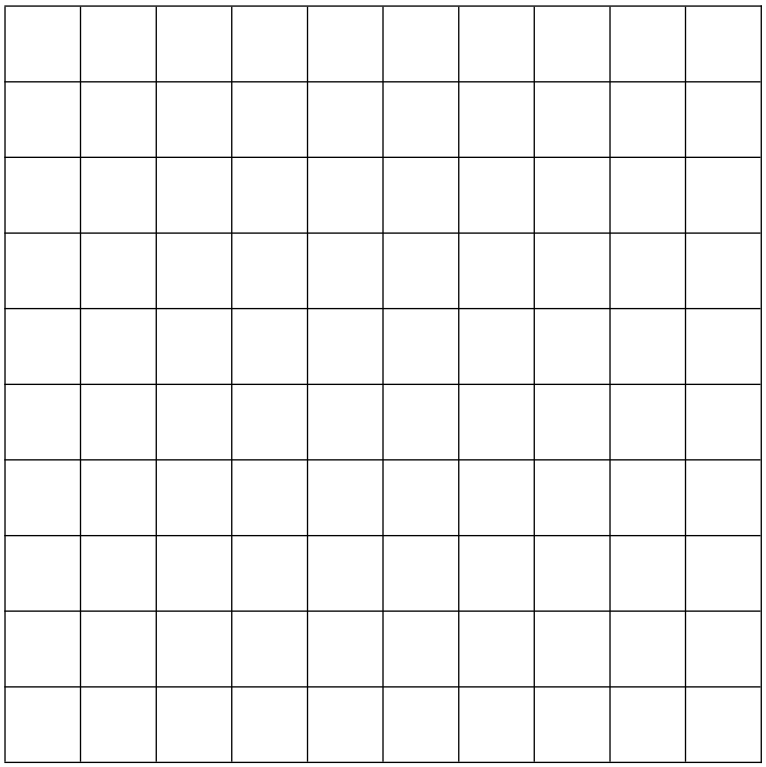

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Select **Next** to display the screen titled ‘How much sugar can dissolve?’

5. Test how many spoonfuls of sugar dissolve at each temperature. Record your data in this table.

| temperature | 20 °C | 40 °C | 60 °C | 80 °C |
|---|---|---|---|---|
| mass of sugar dissolved | | | | |

6. Plot mass of sugar dissolved at different temperatures.



7. How does change in temperature affect the amount of sugar that dissolves?

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Select **Next** to display the screen titled ‘How does temperature affect dissolving?’

8. Watch the particles of water and sugar as the temperature increases. Describe what you see.

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9. What factors do you think affect how quickly sugar dissolves in water, and why?

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10. Select **Next** to display the screen titled '*Getting the solute/solvent out of solution*' and complete the following table.

| MIXTURE TO SEPARATE | SEPARATION METHOD | WHAT HAPPENS TO PARTICLES? |
|---|---|---|
| SAND from seawater | | |
| SUGAR from water | | |
| RED DYE from black dye | | |
| SALT from seawater | | |
| WATER from lemonade | | |


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| | | |
|---|---|---|
| CARBON DIOXIDE from fizzy water | | |

11. All of the substances in question 10 are solutions except for one. Which is not a solution, and why?

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12. What would be the best way to separate salt from a mixture of sand and salt?

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13. What method would you use to separate water from black ink?

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14. Use your knowledge of the properties of solids, liquids and gases to describe how one of the methods of separation works.

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